Terahertz-Induced Second-Harmonic Generation in Quantum Paraelectrics: Hot-Phonon Effect

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Recent THz-pump second-harmonic-generation (SHG) probe measurements of quantum paraelectrics observed a significant long-lived nonoscillatory SHG component following an ultrafast resonant excitation of the soft mode, which was interpreted as a signature of THz-induced transient ferroelectric order. We propose that the THz-induced modulation of the SHG signal can be attributed solely to the dynamic variation of the dielectric environment associated with the lattice background, which reflects the coherent response of soft mode under THz pumping. We develop a temperature-dependent dynamic model incorporating the hot-phonon effect to simulate the soft-mode behaviors under ultrafast THz excitation. Its application to paraelectric $KTaO_3$ produces quantitatively most of the features exhibited in our time-resolved SHG measurements and those in existing literature, including a long-lived nonoscillatory SHG response, SHG oscillations at twice the soft-mode frequency, SHG dampings, and temperature and field-strength dependencies. We conclude that the observed THz-induced nonoscillatory SHG response in quantum paraelectrics is a consequence of the nonequilibrium hot-phonon effect, offering an alternative to its existing interpretation as a signature of transient ferroelectric order.

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Introduction—The quantum criticality in condensed matter physics describes the ordering of a quantum phase that occurs near zero temperature. This phenomenon has attracted considerable attention due to its distinct characteristics arising from the low-lying collective excitations. Extensive research over the past few decades has suggested the presence of quantum criticality in strongly correlated materials, which are often complex with various intertwined quantum orders. A notable exception is the displacive quantum paraelectrics [1–8], where a strong competition between quantum fluctuations and ferroelectric ordering exists. This class of material is supposed to transition from the paraelectric to ferroelectric states at low temperature due to lattice dynamical instability, but the zero-point lattice vibrations prohibits the long-range ferroelectric order [1,5,6,8], leading to an incipient ferroelectricity, sometimes referred to as the "hidden ferroelectric phase" [9,10].

While doping [11–14] or isotope substitution [2,15] can turn quantum paraelectrics to ferroelectrics, transiently reaching the hidden ferroelectric phase through ultrafast manipulation with an intense femtosecond-pulsed laser [16–19] is particularly appealing [20,21]. One strategy is to coherently drive the so-called soft-phonon mode—associated with lattice dynamical instability [22–25]—into the nonlinear regime, aiming to create transient ferroelectric order. Several THz-pumped second-harmonic-generation (SHG) probe measurements have therefore been performed in typical displacive quantum paraelectrics: $SrTiO_3$ [9] and $KTaO_3$ single crystals [10,26]. At low temperatures, a significant long-lived nonoscillatory SHG component, superimposed by a clear SHG oscillation at twice the soft-mode frequency, was observed after a THz pulse [9,10,26]. The origin and precise mechanisms of this nonoscillatory component and soft-mode frequency doubling phenomenon [9,10,26] remain unclear [27,28]. The observed nonoscillatory SHG background is commonly interpreted as a signature of a THz-induced transient ferroelectric order, possibly arising from a THz-driven intrinsic lattice displacement [9], or from a THz-induced long-range correlation between extrinsic local polar structures by defects [10]. However, such an interpretation seems to contradict several other findings. For example, it was realized that the THz pulses up to 500 kV/cm were insufficient to produce a global intrinsic ferroelectricity [10], whereas inducing a long-range correlation between extrinsic defect dipoles via an ultrafast manipulation does not require coherently driving an intrinsic soft mode. More importantly, soft modes are experimentally observed to go through hardening with an increasing THz-field strength [9,10,26], suggesting an intense THz field drives the quantum paraelectrics away from rather than toward ferroelectricity,

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since the soft-mode hardening in displacive paraelectrics is an indicator of departure from ferroelectricity.

On the other hand, similar nonoscillatory components in pump-probe measurements have been commonly observed in other subfields, e.g., in measurements of the interband transitions in semiconductors (e.g., graphene [29]) and collective excitations in superconductors [30–34], and they were attributed to the induced nonequilibrium hot-quasiparticle effect. For example, in superconductors, by using intense THz pulses, one can resonantly excite the amplitude mode of the superconducting order parameter [30–32,35,36], namely the Higgs-mode excitation [37]. In most of the measurements, a nonoscillatory component was observed and can persist for a long time after the THz pulse [30–32], as a consequence of the nonequilibrium hot-quasiparticle effect [30–34]. The hot-quasiparticle effect is a fundamental aspect of nonequilibrium dynamics, applicable to both fermions (electrons) and bosons (phonons) under an external excitation due to the energy input. However, a dedicated hot-phonon effect for purely soft-phononic dynamics under ultrafast optical excitation in ferroelectrics and paraelectrics has largely been overlooked.

In this Letter, we propose that a finite probe-field SHG signal originates from the oxygen-vacancy defects and remains nearly unchanged during the THz excitation. The observed THz-induced SHG modulation can be attributed solely to the dynamic variation of the dielectric environment associated with the lattice background, which reflects the soft mode's coherent response under THz pumping. For this purpose, we develop a temperature-dependent dynamic model to describe the soft-mode behaviors under an ultrafast resonant excitation by incorporating the hot-phonon effect, and examine the THz-induced SHG responses in quantum paraelectric $KTaO_3$ using a combination of theoretical and experimental studies.

Simulations based on our model quantitatively produce all the main features exhibited in our time-resolved SHG measurements, including a long-lived nonoscillatory response, oscillations at twice the soft-mode frequency, SHG dampings, and temperature and field-strength dependence. We therefore attribute the THz-induced nonoscillatory SHG component in quantum paraelectrics to the nonequilibrium hot-phonon effect. Additionally, we also explore the SHG responses of a ferroelectric $KTaO_3$ after an ultrafast THz excitation to understand the individual responses of actual ferroelectric nanoregions (e.g., extrinsic local polar structures by defects [10]). In this case, both our theoretical simulations and experimental measurements show a THz-induced long-lived SHG oscillation at the single polar-mode frequency without any evident signature for the nonoscillatory component, in contrast to the observed THz-induced SHG response of quantum paraelectrics, suggesting that the previously reported resonant SHG features in quantum paraelectrics does not come from ferroelectric nanoregions.

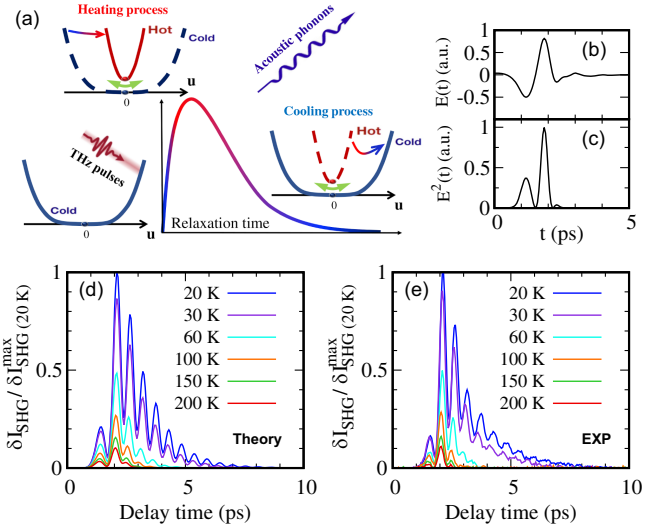

FIG. 1. (a) Schematic illustration of the generation (heating process) and decay (cooling process) of the hot-phonon effect during a nonlinear ultrafast resonant excitation. (b) The waveform and (c) waveform squared of the single-cycle THz pump pulse employed in experiments and simulations. (d) Theoretically calculated and (e) experimentally measured time-resolved SHG in paraelectric $KTaO_3$ at different T.

Experimental setup—We use intense single-cycle THz pump pulses (up to 210 kV/cm), generated from the optical rectification in a $LiNbO_3$ prism [26], to resonantly excite the soft mode in a quantum paraelectric $KTaO_3$ single crystal and detect the time-resolved SHG signal via a femtosecond optical-frequency (800-nm) probe pulse as a function of delay time with respect to the pump pulse. The field directions of the THz pump and the optical-frequency probe pulses are both set along the [100] direction in $KTaO_3$ crystal. The time-resolved SHG signal is measured using a blue filter after the sample, which eliminates the fundamental 800-nm wavelength from the detected light. The waveform of the employed single-cycle THz pump pulses is shown in Fig. 1(b). For our experimental conditions (for example, at mediate 80 K, and a laser spot size around 50 μm focused by a 200 mm lens), the SHG signal read on a lock-in amplifier detected by a GaP photodiode detector is approximately 2.9 μV with 220 μW probe intensity, confirming negligible static contribution. Under similar experimental conditions but 500 μW probe intensity, after resonant excitation by a 210 kV/cm THz field, the reading of the SHG signal reaches up to approximately 120 μV at the peak position in the positive time delay, reflecting a significant (8-times) enhancement beyond the static defect-related background.

Dynamic model—Optical excitation of the soft mode in quantum paraelectrics is known to induce an electrical polarization $\mathbf{P} = u_{sp}(\sum_i Q_i \mathbf{e}_i)/\Omega_{cell}$ [3,38–44], with u_{sp} being the soft-mode displacement; Q_i and $\mathbf{e}_i$ standing for the charges and eigenvectors of the related ions (in a unit cell of volume Ω_{cell}) in the soft mode, respectively. As a

result of lattice dynamics, its effective Lagrangian can be written as

$$\mathcal{L}_{\text{eff}} = \frac{m_p}{2}(\partial_t P)^2 - \left[\frac{\alpha(T)}{2}P^2 + \frac{b}{4}P^4 - \mathbf{E}(t)\cdot\mathbf{P}\right], \quad (1)$$

where b is an anharmonic coefficient, related to three-phonon interactions; m_p denotes the effective mass [38]; $\mathbf{E}(t)$ represents the THz field; $\alpha(T)$ is the harmonic coefficient, and using the self-consistent renormalization theory within the path-integral approach [6], the equilibrium harmonic coefficient is derived as $\alpha_e(T) = \alpha_e(T = 0) + bC(T)$ [see Supplemental Material (SM) [45]]. Here, $C(T)$ requires a self-consistent formulation of the bosonic thermal excitation of the soft phonons,

$$C(T) = \frac{\hbar}{m_p}\sum_{\mathbf{q}}\left[\frac{2n_q^{(0)}+1}{\omega_q(T)} - \frac{1}{\omega_q(T=0)}\right], \quad (2)$$

where $n_q^{(0)} = 1/\{\exp[\hbar\omega_q/(k_B T)] - 1\}$ is the equilibrium distribution function (Bose distribution) of the soft phonons, and the energy spectrum of the soft phonons is given by

$$\omega_q(T) = \sqrt{[\alpha_e(T=0) + bC(T)]/m_p + v^2q^2}, \quad (3)$$

with v being the mode velocity. Mathematically, $C(T)$ in Eq. (2) increases monotonically with temperature, and as seen from Eq. (3), this monotonic increase describes the soft-mode hardening [i.e., the increase of soft-phonon excitation gap $\omega_{q=0}(T)$] with temperature, consistent with the known behavior of the soft modes in quantum paraelectrics [22–25].

The ultrafast THz field $\mathbf{E}(t)$ can stimulate the dynamics of $\mathbf{P}(t) = \mathbf{P}_0 + \delta\mathbf{P}(t)$, and in particular, a nonequilibrium distribution $n_q = n_q^{(0)} + \delta n_q(t)$ of the soft phonons, thereby leading to the evolution of $\alpha(t) = \alpha_e + \delta\alpha(t)$ according to Eq. (2). Using the Euler-Lagrange equation [60], one can find the equation of motion for the polarization,

$$m_p\partial_t^2\mathbf{P} + \gamma\partial_t\mathbf{P} = -\alpha(t)\mathbf{P} - bP^2\mathbf{P} + \mathbf{E}(t). \quad (4)$$

Here, we have introduced a damping term, with γ being the damping rate. This damping should be dominated by three-phonon scattering between two soft phonons and one acoustic phonon, leading to a T-dependent $\gamma(T)$. In principle, the evolution of $\alpha(t)$ should incorporate a fully microscopic bosonic Boltzmann equation of the soft phonons. However, such a microscopic treatment is complex and will not change the main results and conclusions in this Letter. Thus, here we employ the Allen-Cahn-like relaxation equation extensively used in the phase-field method [46] (see SM [45]),

$$\partial_t\delta\alpha(t) = -\eta(P^2 - P_0^2)/2 - \delta\alpha/\tau_E, \quad (5)$$

where we take $\eta < 0$ since the soft-phonon number increases after the excitation and leads to the increase of $\alpha(t)$ according to Eq. (2); τ_E is the energy-relaxation time of the system. The physical picture of hot-phonon effect is shown in Fig. 1(a). The initial THz field induces a hot-phonon effect due to the energy input, which manifests itself as the increase of the harmonic coefficient of the potential well, leading to an increased nonoscillatory SHG component in the first picosecond and hence the hardening of the soft-mode oscillation (green arrow). This effect gradually decays after the THz field due to cooling via interactions with acoustic phonons, leading to exponential decay of the nonoscillatory SHG component.

For the simulation in quantum paraelectrics, we first self-consistently solve the coupled Eqs. (2) and (3) using only the knowledge about the ground-state parameters $\alpha_e(T = 0)$ and b, to obtain the equilibrium $\alpha_e(T) > 0$, and hence, $P_0^2 \equiv 0$. We then solve the dynamic equations [Eqs. (4) and (5)] using experimental waveform of the single-cycle THz pump pulse [Fig. 1(b)] as the input field $\mathbf{E}(t)$, resulting in the temperature-dependent soft-mode dynamics under an ultrafast excitation.

The inverse dielectric function $1/\varepsilon(T) \propto \alpha(T)$ in quantum paraelectrics [2,6], leading to the THz-induced SHG intensity $\delta\mathrm{I}_{\mathrm{SHG}}(t) \propto \delta\alpha(t)$ (see SM [45]). In specific simulations, we consider the damping rate $\gamma(T)$ as the only fitting parameter for temperature variation in the experimental measurements. Other parameters used in simulations are based on several independent experimental measurements.

Results—Figures 1(d) and 1(e) show the theoretically predicted and experimentally measured time-resolved SHG responses under THz excitation at different temperatures, respectively. At a low temperature of 20 K, the THz field can coherently drive the soft mode into a strong nonlinear resonant state, showing clear oscillations on top of a nonoscillatory background after the THz stimulation ($t > 2$ ps). These resonant features can persist up to 8 ps at 20 K, and gradually ebb away as temperature increases. At a high temperature of 200 K, the SHG response becomes weak and only follows the square of the THz waveform [Fig. 1(b)] because the soft mode moves out of the range of the THz spectrum due to its hardening with increase in temperature. All of these THz-induced SHG features are consistent with the previously reported findings in quantum paraelectrics $SrTiO_3$ [9] and $KTaO_3$ [10,26].

For a direct comparison, we plot the theoretical and experimental results together in Fig. 2(a), which demonstrates a remarkably quantitative agreement between our theoretical predictions and experimental measurements in all aspects of the THz-induced SHG responses of the quantum paraelectric. It should be emphasized that after the zero-temperature parameters are determined from

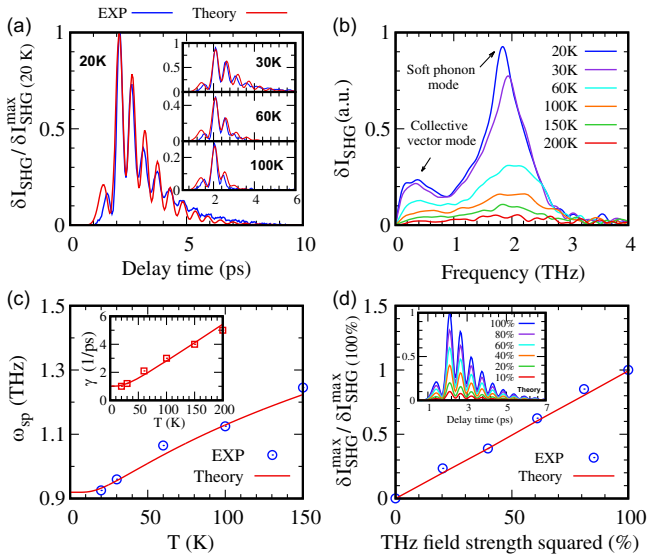

FIG. 2. (a) Detailed comparison between theoretical and experimental results in Figs. 1(d) and 1(e). (b) Fourier transform of the measured SHG oscillations at different T. At temperatures below 100 K, besides the widely reported soft mode that emerges around 1.8 THz, there emerges another mode around lower frequency of 0.4 THz, which should correspond to a collective vector mode emerged in displacive ferroelectrics and quantum paraelectrics as proposed in Ref. [6]. (c) Theoretical and experimental results of the soft-mode excitation gap. The inset shows the employed damping rate $\gamma(T)$ (squares) in our simulation, and a fitting curve mentioned in main text with $T_{\rm ac} = 76$ K. (d) SHG peaks versus THz-field power at 20 K. The inset shows the numerical time-resolved SHG signal at different field strengths.

independent measurements, our simulation achieves this good agreement in both temperature variation and temporal evolution by fitting only a single parameter, the temperature-dependent damping rate $\gamma(T)$.

To gain more insight into these emerging SHG characters of quantum paraelectrics under a THz excitation, we carefully examine the temperature-dependent behaviors of the soft-mode hardening and the SHG damping. Figure 2(b) shows the FFT of the measured SHG oscillatory component, which was acquired by subtracting the nonoscillatory SHG component from the original signal. At 20 K, a single mode emerges around 1.8 THz, and it gradually hardens as temperature increases. This is consistent with the established soft-mode behavior of $KTaO_3$ reported in previous studies [10,26]. It should be emphasized that the soft mode in quantum paraelectric $KTaO_3$ is reported to downshift to 0.8–0.9 THz below 50 K [22–25]. Thus, the THz-induced SHG oscillations in both our theory and experiment oscillate at twice the soft-mode frequency. This soft-mode frequency doubling phenomenon [9,10,26] is expected in quantum paraelectrics lying at the verge of the central-symmetric state (we will discuss its origin later). The extracted soft-mode frequencies from experiments and the simulations [$\omega_{q=0} = \sqrt{\alpha_e(T)/m_p}$] are plotted together in Fig. 2(c) as a function of temperature and they exhibit an excellent quantitative agreement in their temperature dependence.

For an analytical analysis of the THz excitation, we assume a pump field in the single-frequency form $\mathbf{E}(t) \approx \mathbf{E}_0 \cos(\Omega t)$, with Ω being the THz-field frequency. From Eqs. (4) and (5), neglecting all damping terms, one approximately has $\partial_t \delta\alpha(t) \approx (|\eta| E_0^2 \cos^2(\Omega t)/2[m_p(\omega_{q=0}^2 - \Omega^2)]^2)$ (see SM [45]), leading to the THz-induced dynamic behavior of $\delta\alpha(t)$,

$$\delta\alpha(t) \approx \frac{|\eta| E_0^2 \sin(2\Omega t)/(2\Omega)}{m_p^2[(2\omega_{q=0})^2 - (2\Omega)^2]^2} + \frac{|\eta| E_0^2 t}{m_p^2[(2\omega_{q=0})^2 - (2\Omega)^2]^2}. \tag{6}$$

The first term in the right-hand side of the above equation contributes to the SHG oscillations, induced by the second order of the THz pump field. This second-order response to a THz field leads to the aforementioned soft-mode frequency doubling phenomenon, since for an ultrafast (δ-function) pulse, it becomes $|\eta| E_0^2 \sin(2\omega_{q=0} t)/(8 m_p^2 \omega_{q=0}^2)$ in the response theory of a resonant excitation. The second term leads to the generation of a nonoscillatory SHG component, and it describes the nonequilibrium hot-phonon effect under the nonlinear excitation because it pushes the total $\alpha(t)$ toward a higher positive value, proportional to the power $E_0^2 \sigma$ of the THz pulse, with σ being the THz-pulse temporal width. In addition, by Eq. (6), one can also infer that a significant nonoscillatory SHG component and large SHG oscillations are possible only at the resonant-excitation condition of $\omega_{q=0}(T) = \Omega$, in agreement with our numerical simulations and experimental observations in the temperature-dependent SHG signal [Figs. 1(d) and 1(e)].

Our numerical and experimental results for the SHG signal peaks as a function of the THz-field power are plotted in Fig. 2(d). They are proportional to the THz-field power, i.e., $\delta I_{\rm SHG} \propto E_{\rm pump}^2 E_{\rm probe}^4$, as also observed in previous measurements [9,10,26] and in agreement with the analysis above. This suggests that the THz-induced SHG response in quantum paraelectrics is a second-order response to the THz field.

For the damping, there are two distinct relaxation processes going on after an ultrafast excitation: soft-mode damping $\gamma(T)$ and energy relaxation $1/\tau_E$. We find that the soft-mode damping dominates the damping of the THz-induced resonant SHG features as it should, and it can be well described by the three-phonon (two soft phonons and one acoustic phonon) scattering mentioned above. Specifically, according to the microscopic scattering mechanism for calculating the scattering probabilities of the acoustic-phonon emission and absorption [61,62], the damping rate of the soft mode can be approximated as $\gamma(T) \propto 2\bar{n}_{\rm ac}(T) + 1 \approx (2/e^{T_{\rm ac}/T} - 1) + 1$, with $\bar{n}_{\rm ac}(T)$

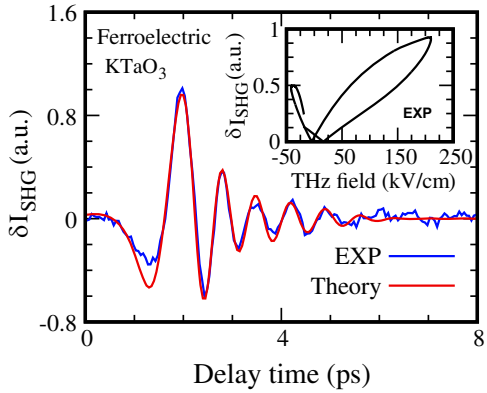

FIG. 3. Numerical simulation and experimental measurement of the time-resolved SHG signal in ferroelectric $KTaO_3$ at 77 K. The inset shows the measured hysteresis loop of this sample at 77 K.

being the averaged acoustic-phonon number and T_{ac} being a characteristic temperature. Then, the temperature dependence of $\gamma(T)$, obtained by fitting to our time-resolved experiments, can be well captured, as illustrated in the inset of Fig. 2(c).

Finally, it is noted that to explain the observed long-lived nonoscillatory SHG component, Cheng *et al.* have proposed a potential origin in Ref. [10], a THz-induced correlation between the local polar structures (i.e., ferroelectric nanoregions) that arise from the extrinsic defects, leading to a global ferroelectriclike response. To examine this possibility, we explore the THz-induced SHG response of a ferroelectric $KTaO_3$, experimentally obtained through an annealing process [63]. We then conducted the THz-pump SHG-probe measurements on the ferroelectric $KTaO_3$ crystal, and the results are plotted in Fig. 3. As shown in Fig. 3, the ferroelectric $KTaO_3$ exhibits totally different THz-induced SHG responses from the quantum paraelectric $KTaO_3$. The THz-field strength dependence of the SHG response (inset of Fig. 3) shows a butterfly-shape hysteresis loop of the THz-induced SHG change [69], suggesting the presence of the ferroelectricity. As for the time-resolved SHG signal (Fig. 3), a clear oscillation develops after the THz pulse ($t > 2$ ps) and persists up to 7 ps, suggesting a coherent or resonant driving of a collective excitation (i.e., a polar mode). However, no evident nonoscillatory component was observed, and no polar-mode frequency doubling phenomenon occurs in the resonant excitation as the observed SHG response during the pump pulse ($t < 2$ ps) just follows the pump-pulse waveform [Fig. 1(b)]. This suggests that the THz-induced SHG signal in ferroelectric $KTaO_3$ is a linear response to the THz pump field, i.e., $\delta I_{SHG} \propto E_{pump} E_{probe}^4$, as a consequence of the breaking of the global lattice inversion symmetry by the existing ferroelectric order. These resonant features in the ferroelectric state are in contrast to the ones in the quantum paraelectric state, and therefore, it should suggest that the ferroelectric nanoregions are not the origin for the observed SHG resonant features in the quantum paraelectrics.

For numerical simulations of the ferroelectric state, we set a negative value for α_e at $T = 77$ K to fit our experimental measurement of ferroelectric $KTaO_3$, and it leads to a finite equilibrium $P_0^2 = -\alpha_e/b$. Then, we perform the simulation on the basis of the dynamic model, and as shown in Fig. 3, the produced results can well capture the experimental measurements. Following the derivation of Eq. (6), the THz-induced dynamic behavior of $\delta\alpha(t)$ in the ferroelectric $KTaO_3$ reads

$$\delta\alpha \approx |\eta| \frac{(\mathbf{P}_0 \cdot \mathbf{E}_0)\sin(\Omega t)}{m_p \Omega(\Delta^2 - \Omega^2)} + \frac{|\eta| E_0^2 [\sin(2\Omega t)/(2\Omega) + t]}{4\Omega[m_p(\Delta^2 - \Omega^2)]^2}, \quad (7)$$

with $\Delta = \sqrt{-2\alpha_e/m_p}$ being the polar-mode excitation gap. The first term of this expression corresponds to the linear response of the ferroelectric order to the THz field. The presence of the ferroelectric order leads to the emergence of a linear response to the THz field, and it dominates the THz-induced SHG signal at a relatively weak THz field, in agreement with our numerical simulations and symmetry analysis as well as our observations in experimental measurements on annealed, ferroelectric $KTaO_3$ here, where the SHG intensity follows the THz pulse waveform and scales linearly with the pump field. The hot-phonon effect (nonoscillatory component) and soft-mode frequency doubling that manifest in second-order excitation (observed in quantum paraelectrics) are masked by the dominant linear response, which is why these features are not observed in our measurements on ferroelectric $KTaO_3$, leading to the appearance of polar-mode oscillations, a linear dependence of the THz-induced SHG on pump field, and the absence of a nonoscillatory SHG component unless a significant high-order excitation by strongly intense field is present. Actually, similar phenomena have been widely observed in pump-probe measurements of other ferroelectric materials [66–68,70], including the hysteresis loop of the THz-induced SHG change [67,70].

In summary, combining numerical simulations and experimental measurements on the THz-induced time-resolved SHG responses in paraelectric $KTaO_3$, we conclude that the observed long-lived nonoscillatory component in previous experiments [9,10] is a result of a nonequilibrium hot-phonon effect. Then, the observed soft-mode hardening as THz-field strength increases [9,10,26] can also be understood: the increase in the field strength enhances the hot-phonon effect and promotes the soft-phonon temperature during the nonequilibrium process, thereby leading to a soft-mode hardening.

Previous experiments [9,10] assume that a finite SHG signal necessarily indicates the emergence of ferroelectric order, which breaks inversion symmetry. However, we

propose and experimentally demonstrate that while a finite SHG signal indicates the inversion symmetry breaking, it does not necessarily imply the presence of ferroelectric order. Our measurements (refer to SM [45]) find that increasing oxygen vacancies in paraelectric $KTaO_3$ can significantly enhance the SHG intensity (even without the THz pump) [71], while no signs of ferroelectric order are present in the system. This indicates that a finite probe-field SHG polarization can exist in quantum paraelectrics due to inevitable oxygen vacancies. It most likely arises from the two-photon interband transitions of electrons, mediated by the electronic defect states associated with the oxygen vacancies, which break translational symmetry and locally break inversion symmetry. This suggests that a finite SHG detected by the 800-nm probe field *cannot* be directly used to justify the emergence of ferroelectric order. We therefore call for a careful examination of this signal.

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F. Y. and X. J. L. did the theoretical and experimental studies, respectively, and contributed equally to this Letter.

Data availability—The data that support the findings of this Letter are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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